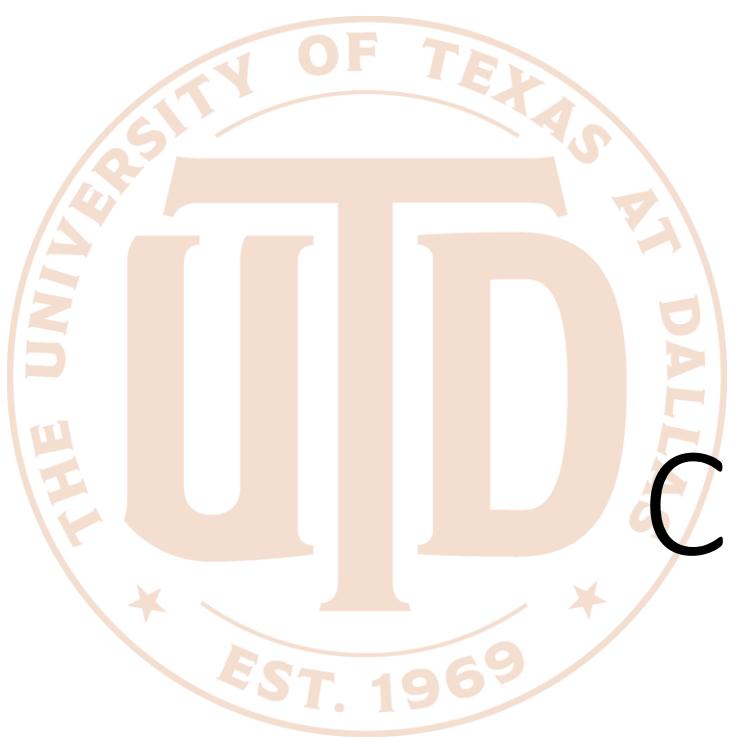




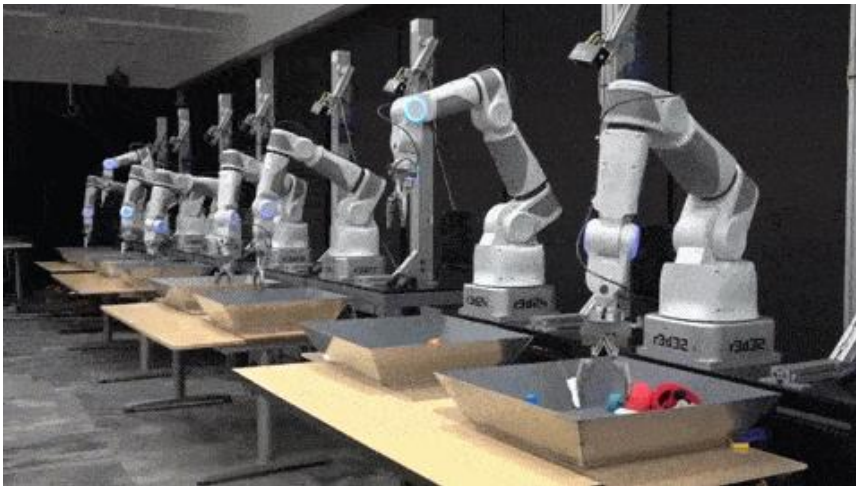
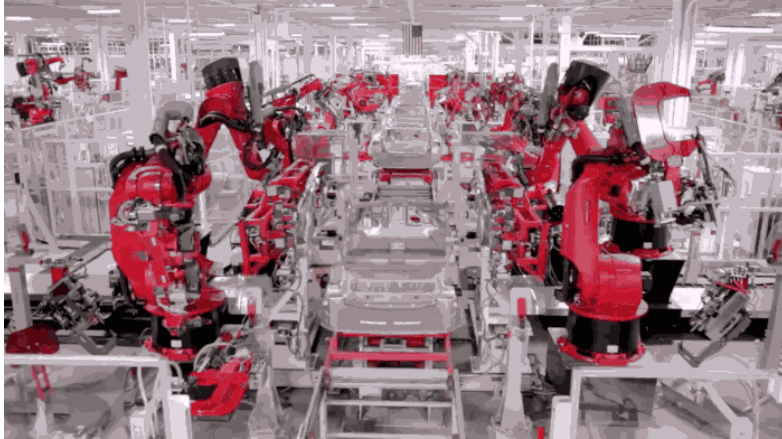
Configuration Space

CS 6341 Robotics

Instructor Yu Xiang

The University of Texas at Dallas

Robotics

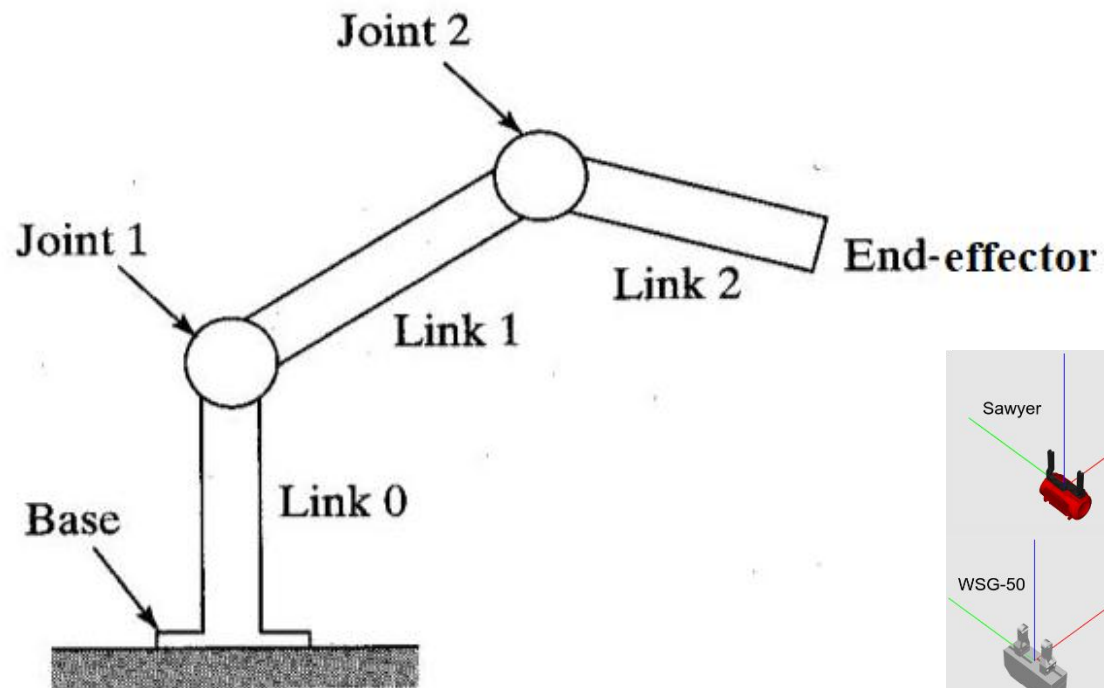


What is the common phenomenon in these robots?

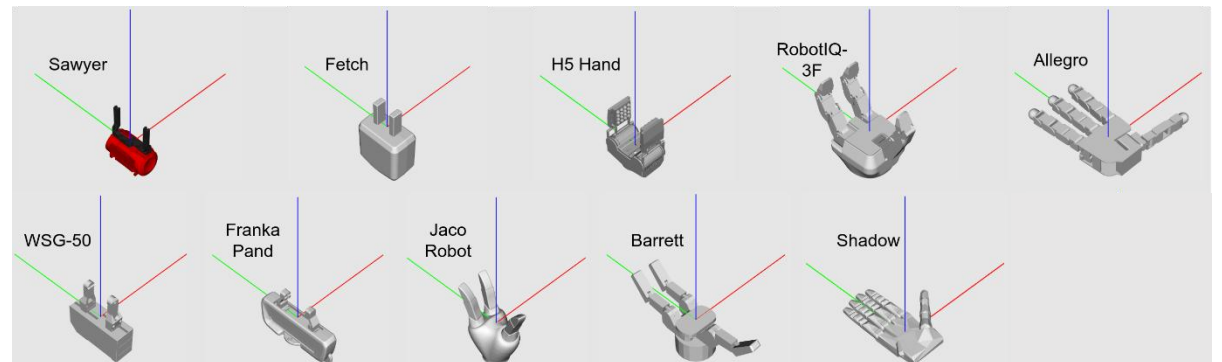
Motion

Robot Mechanisms

- Links and Joints

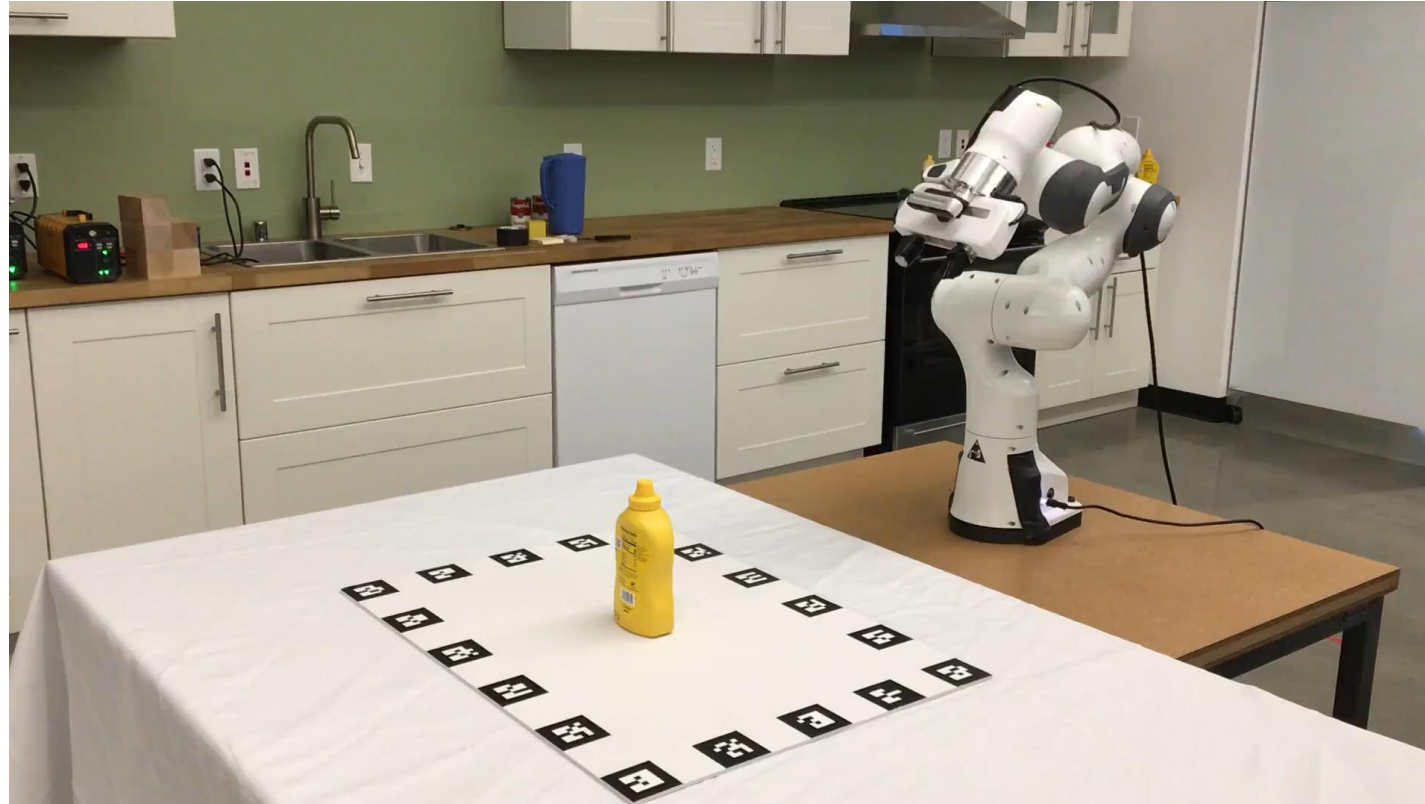


Franka Emika



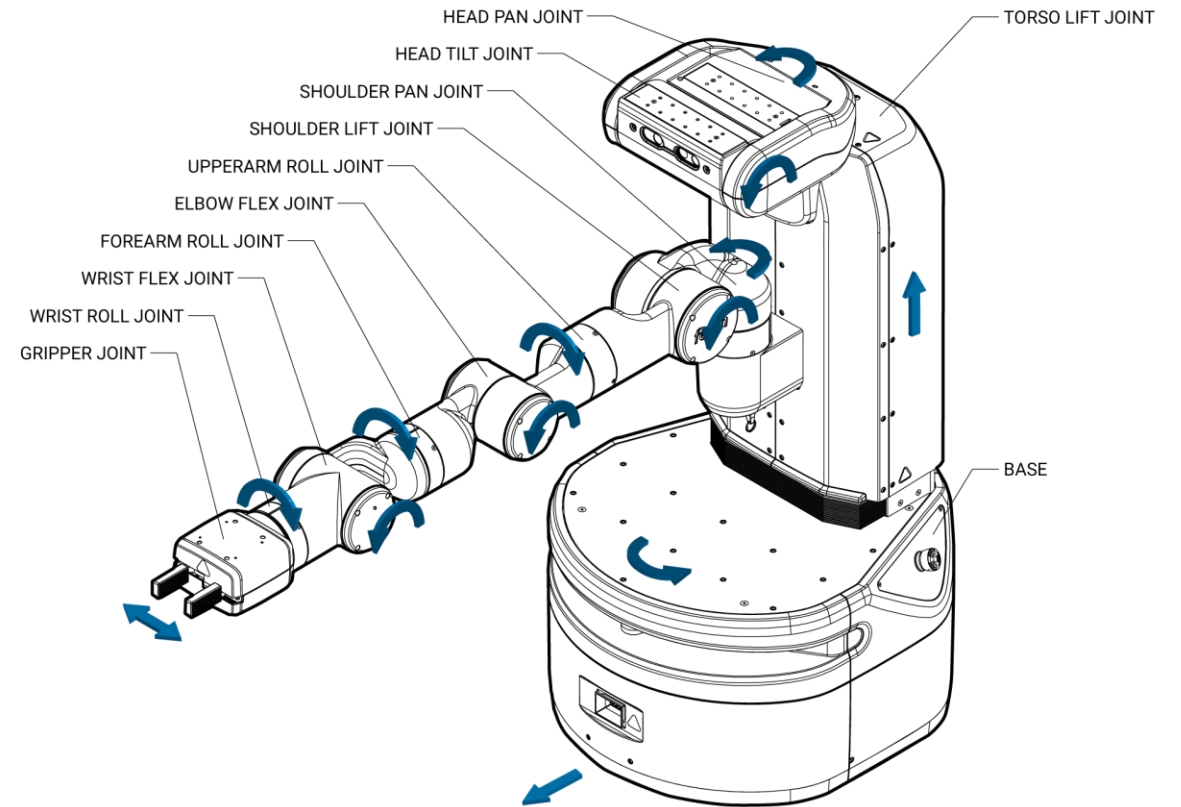
End-effectors

Franka Movement



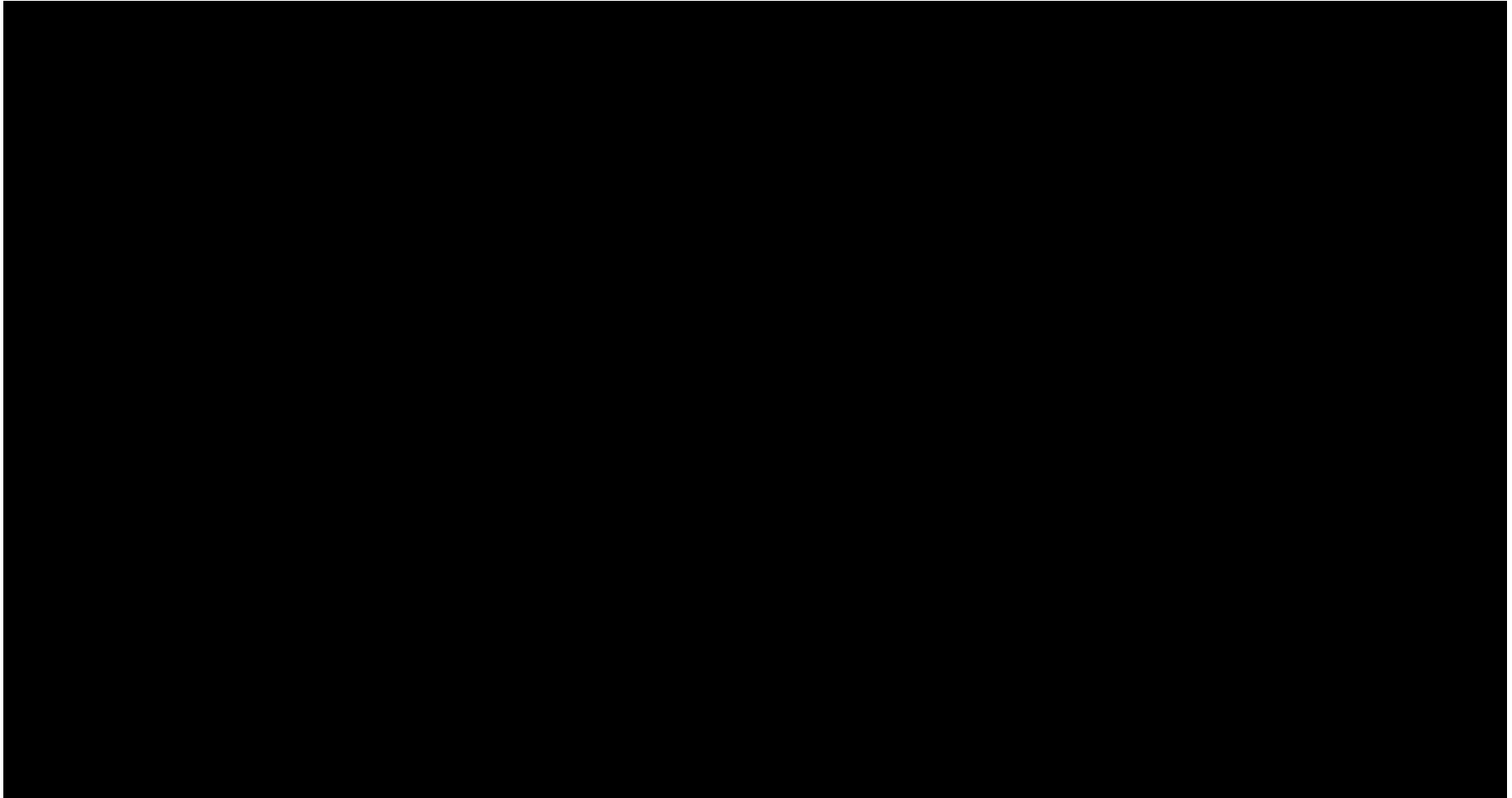
Robot Mechanisms

- Links and Joints



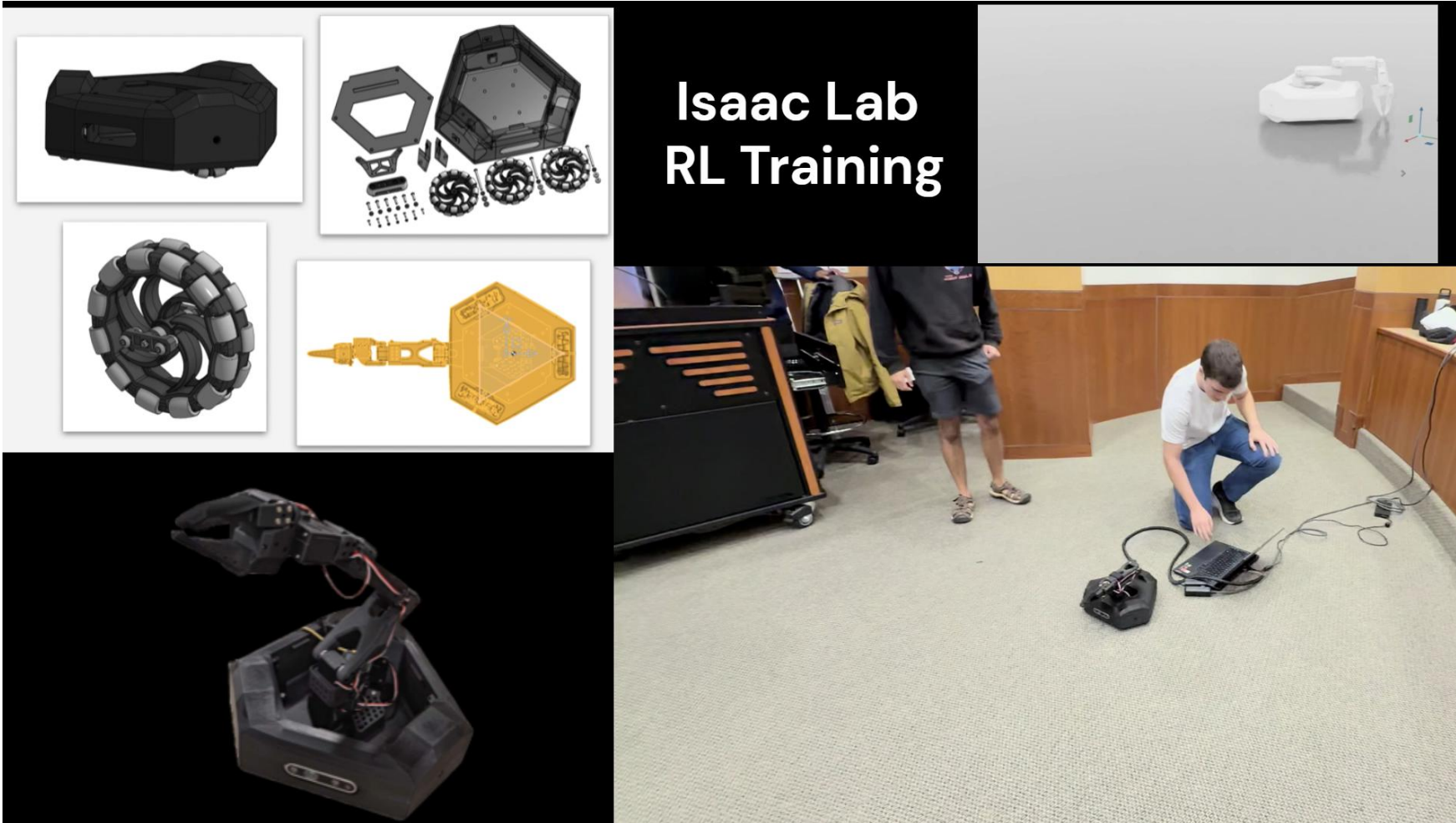
Fetch Mobile Manipulator

Fetch Movement



<https://irvlutd.github.io/SelfSupervisedSegmentation/>

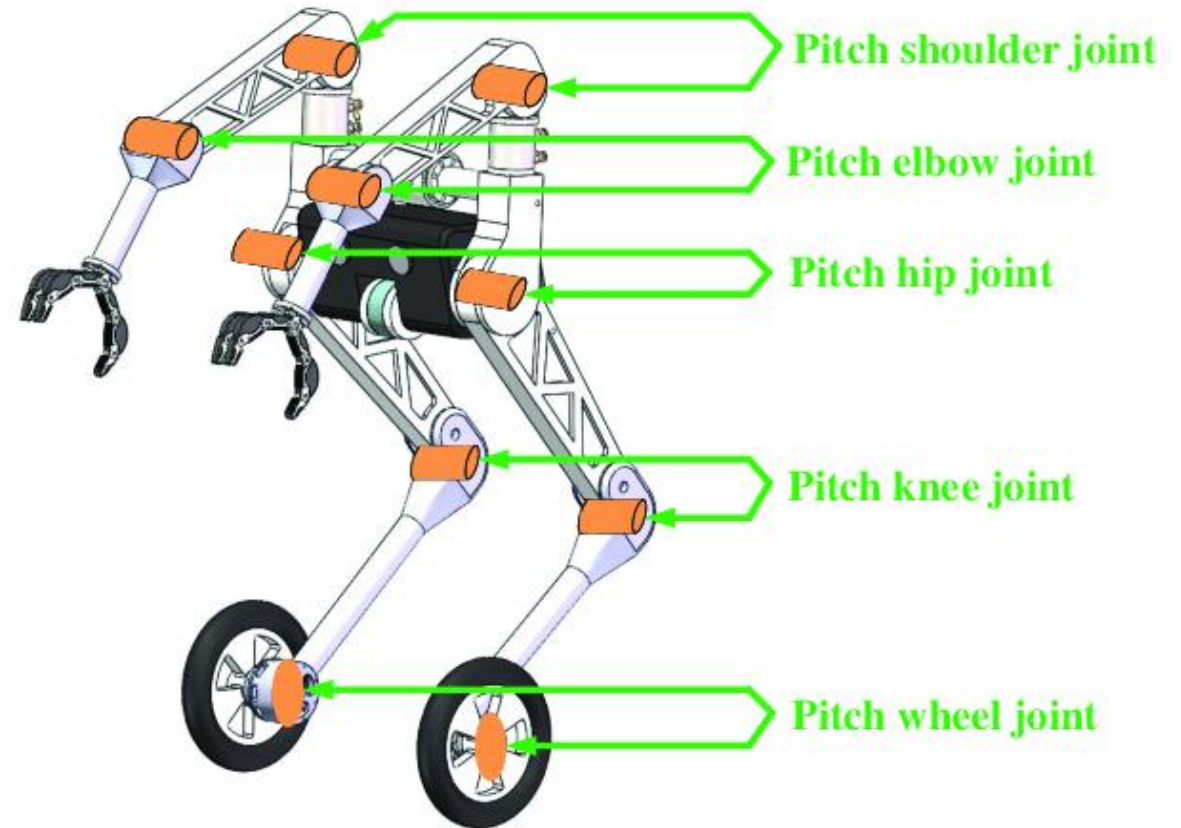
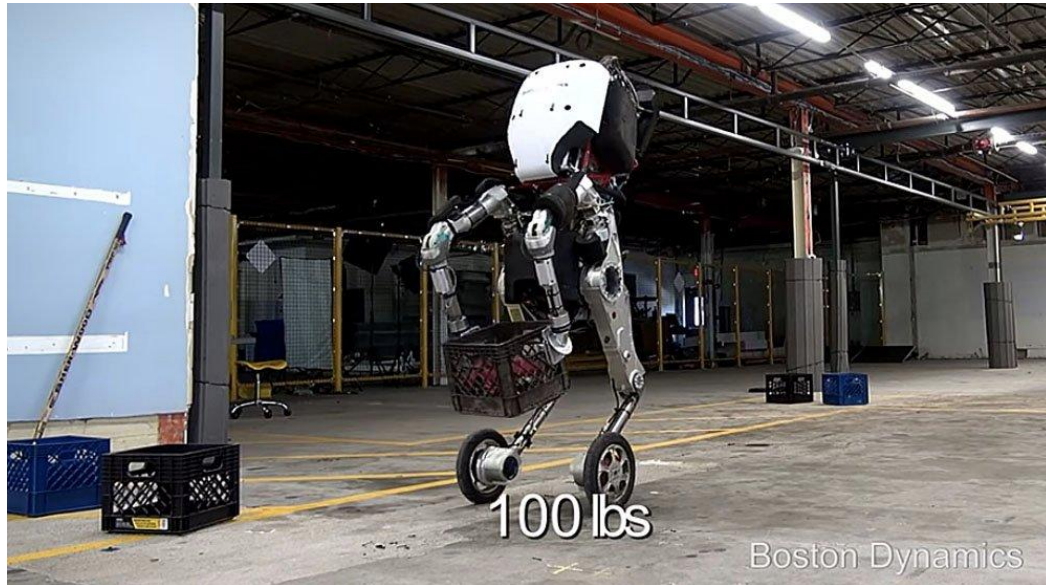
SO101 Movement



Fall 2025 Course Project

Robot Mechanisms

- Links and Joints



<https://thenewstack.io/boston-dynamics-agile-wheel-legged-humanoid-robot-performs-incredible-stunts/>

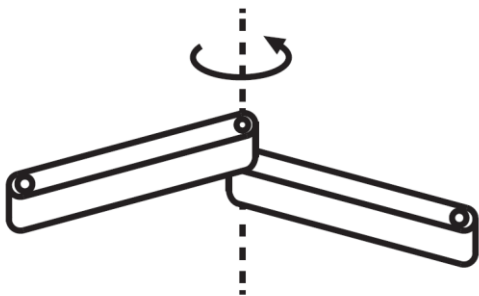
$\pi_{0.5}$: a VLA with Open-World Generalization



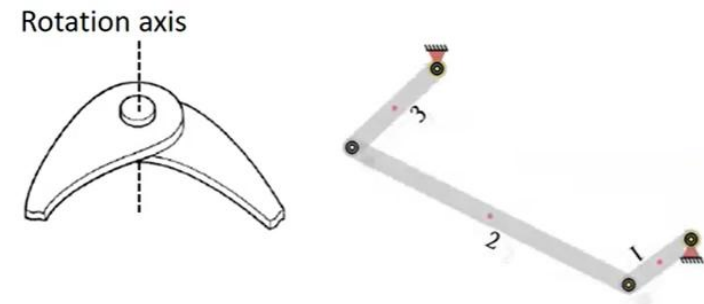
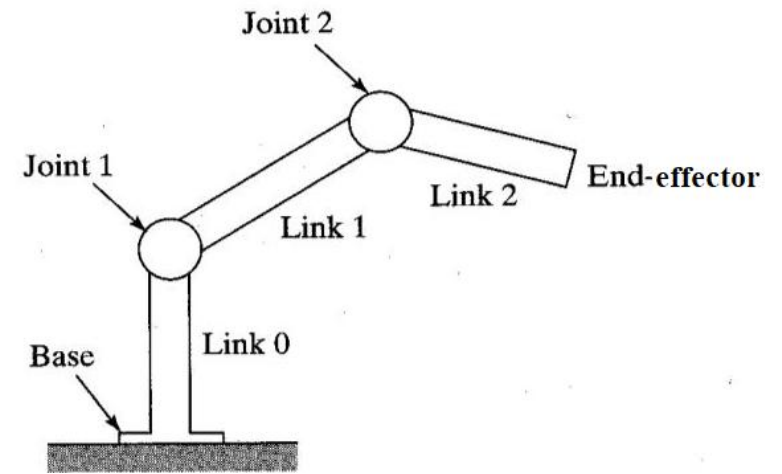
<https://www.physicalintelligence.company/blog/pi05>

Robot Joints

- Every joint connects exactly two links
- Revolute joint (R)
 - Hinge joint
 - Allows rotation motion about the joint axis

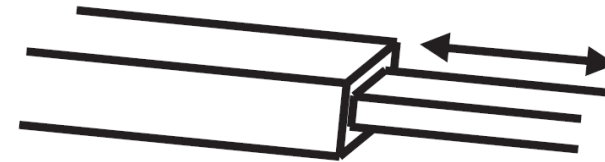


Revolute
(R)

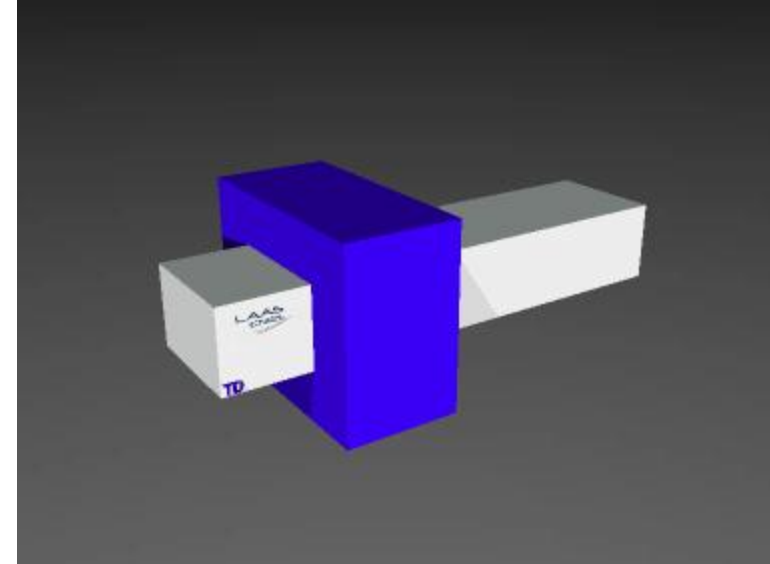


Robot Joints

- Prismatic Joint (P)
 - Sliding joint or linear joint
 - Allows translational motion along the direction of the joint axis

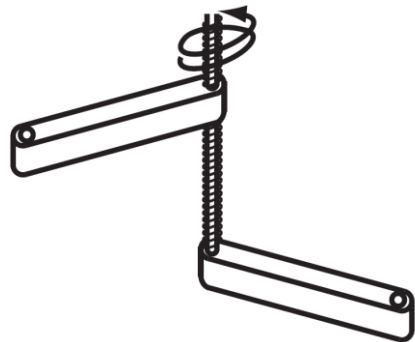


Prismatic
(P)

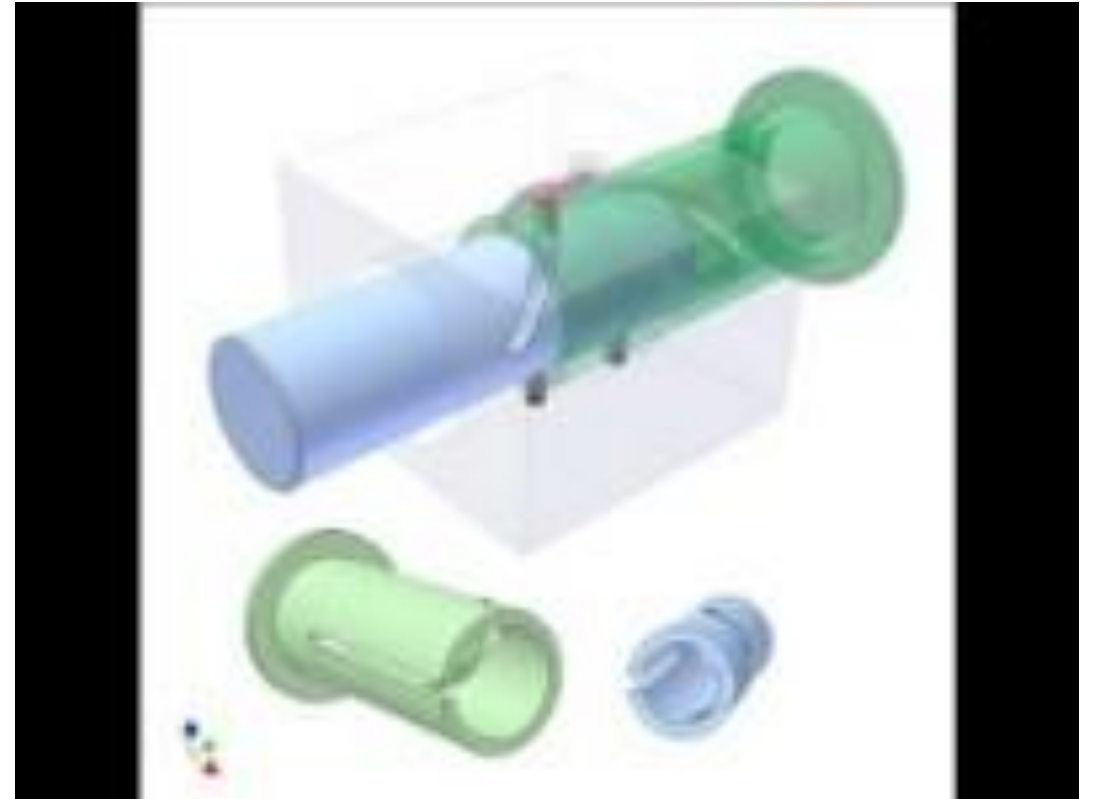


Robot Joints

- Helical Joint (H)
 - Screw joint
 - Allows rotation and translation about a screw axis

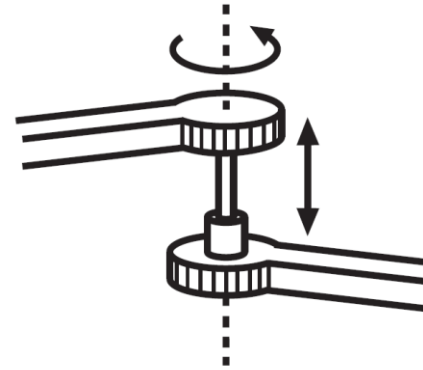


Helical
(H)

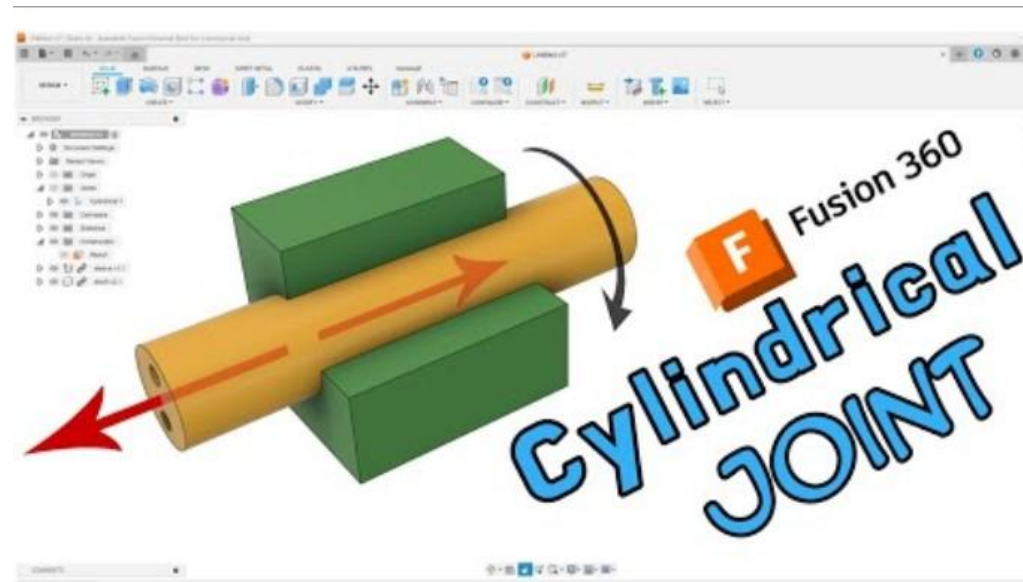


Robot Joints

- Cylindrical joint (C)
 - Allows independent translations and rotations about a single fixed joint axis

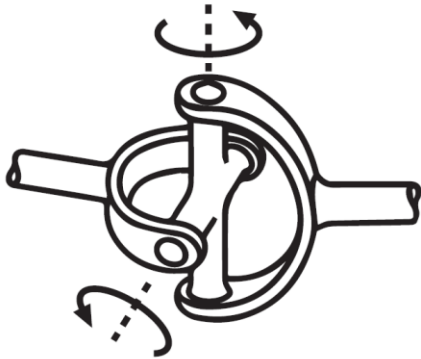


Cylindrical
(C)

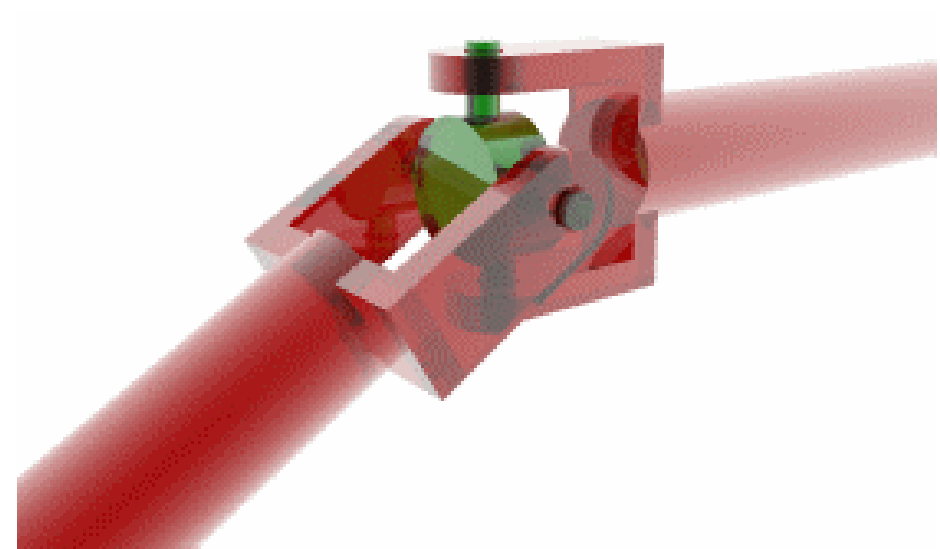


Robot Joints

- Universal joint (U)
 - A pair of revolute joints with orthogonal joint axes

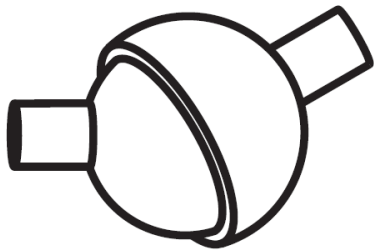


Universal
(U)



Robot Joints

- Spherical joint (S)
 - Ball-and-socket joint



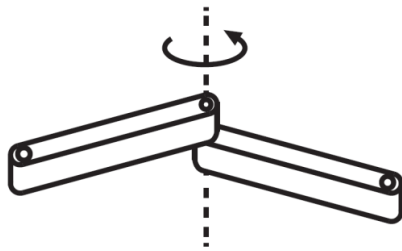
Spherical
(S)



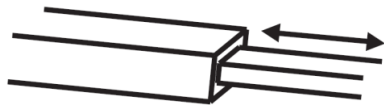
<https://youtu.be/kztZu3uTyvM>

Robot Joints

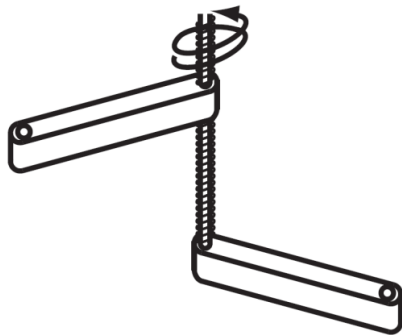
- **Every joint connects exactly two links**



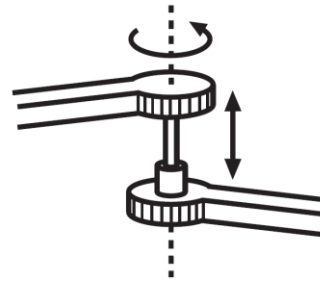
Revolute
(R)



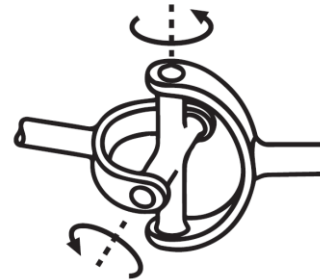
Prismatic
(P)



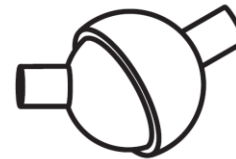
Helical
(H)



Cylindrical
(C)



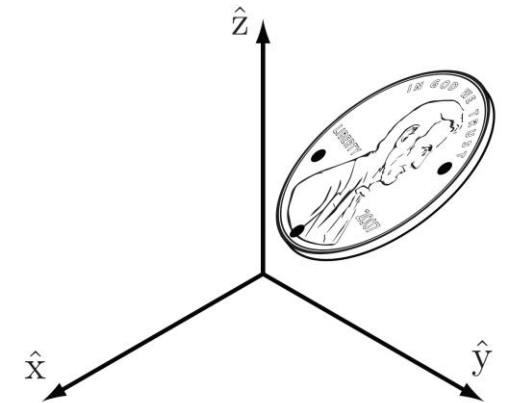
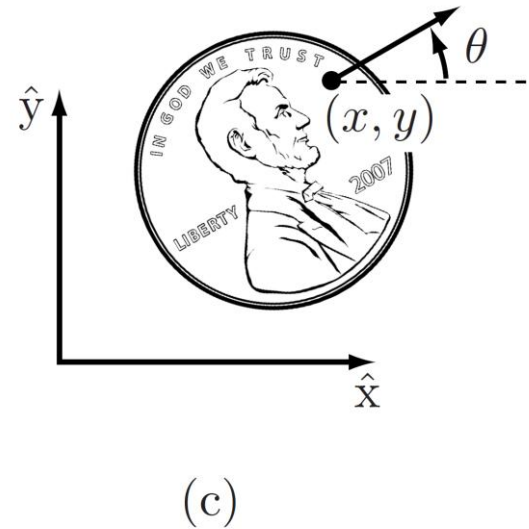
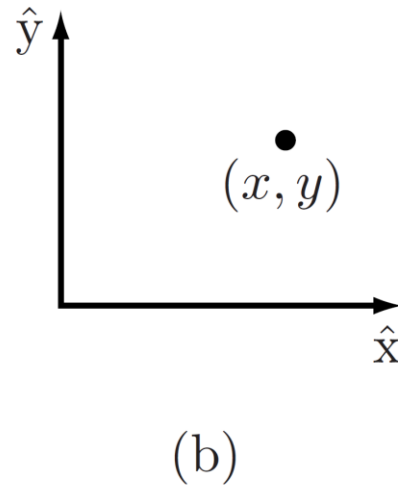
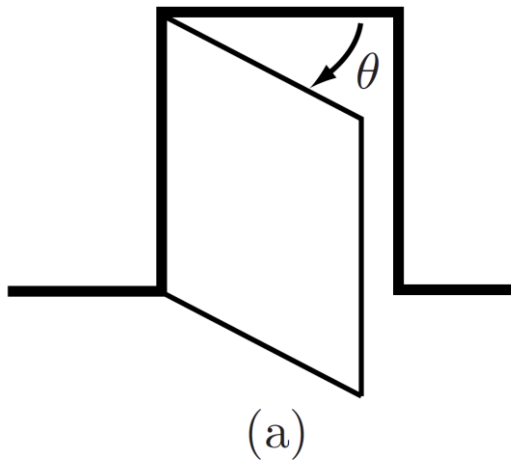
Universal
(U)



Spherical
(S)

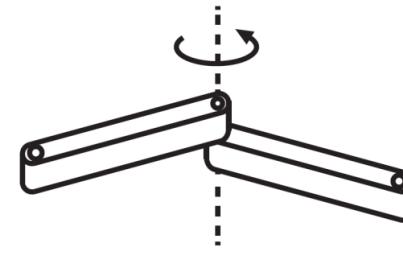
Degrees of Freedom

- Maximum number of logically independent values
- Specify the position of a rigid body



Degrees of Freedom of Robot Joints

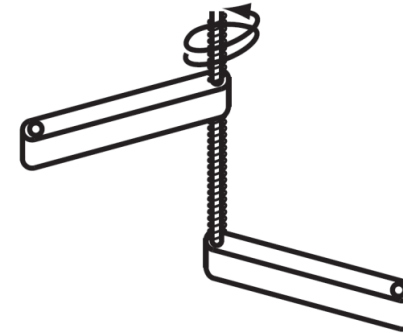
- Revolute joint
 - 1 DOF
- Prismatic joint
 - 1 DOF
- Helical joint
 - 1 DOF



Revolute
(R)



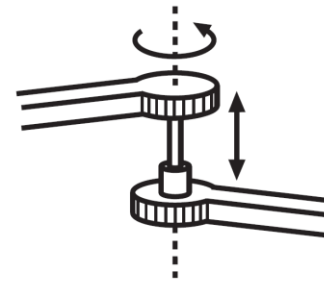
Prismatic
(P)



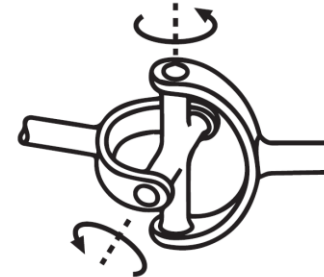
Helical
(H)

Degrees of Freedom of Robot Joints

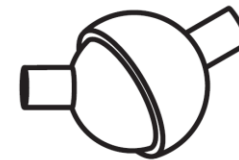
- Cylindrical joint
 - 2 DOF
- Universal joint
 - 2 DOF
- Spherical joint
 - 3 DOF



Cylindrical
(C)

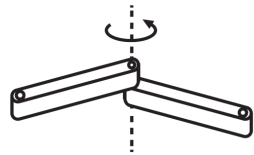


Universal
(U)

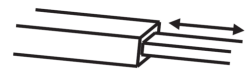


Spherical
(S)

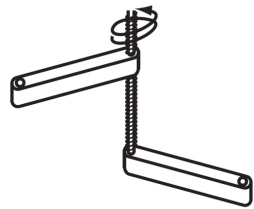
Degrees of Freedom of Robot Joints



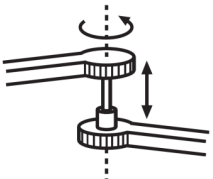
Revolute
(R)



Prismatic
(P)



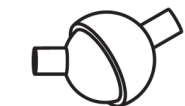
Helical
(H)



Cylindrical
(C)



Universal
(U)



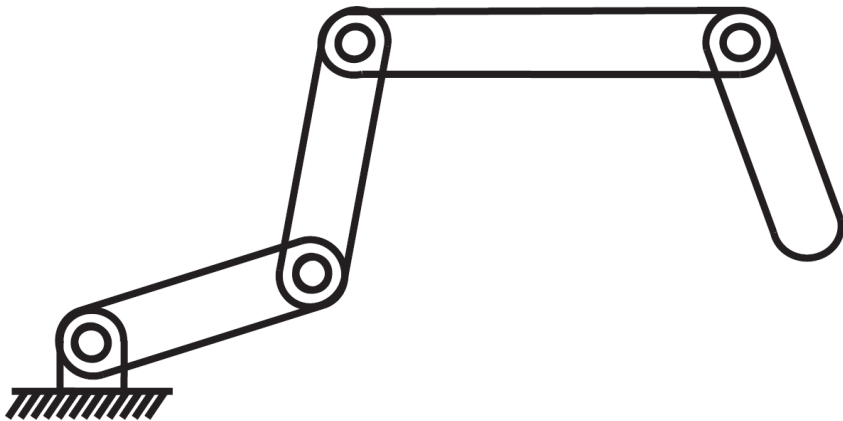
Spherical
(S)

Joint type	dof f	Constraints c between two planar rigid bodies	Constraints c between two spatial rigid bodies
Revolute (R)	1	2	5
Prismatic (P)	1	2	5
Helical (H)	1	N/A	5
Cylindrical (C)	2	N/A	4
Universal (U)	2	N/A	4
Spherical (S)	3	N/A	3

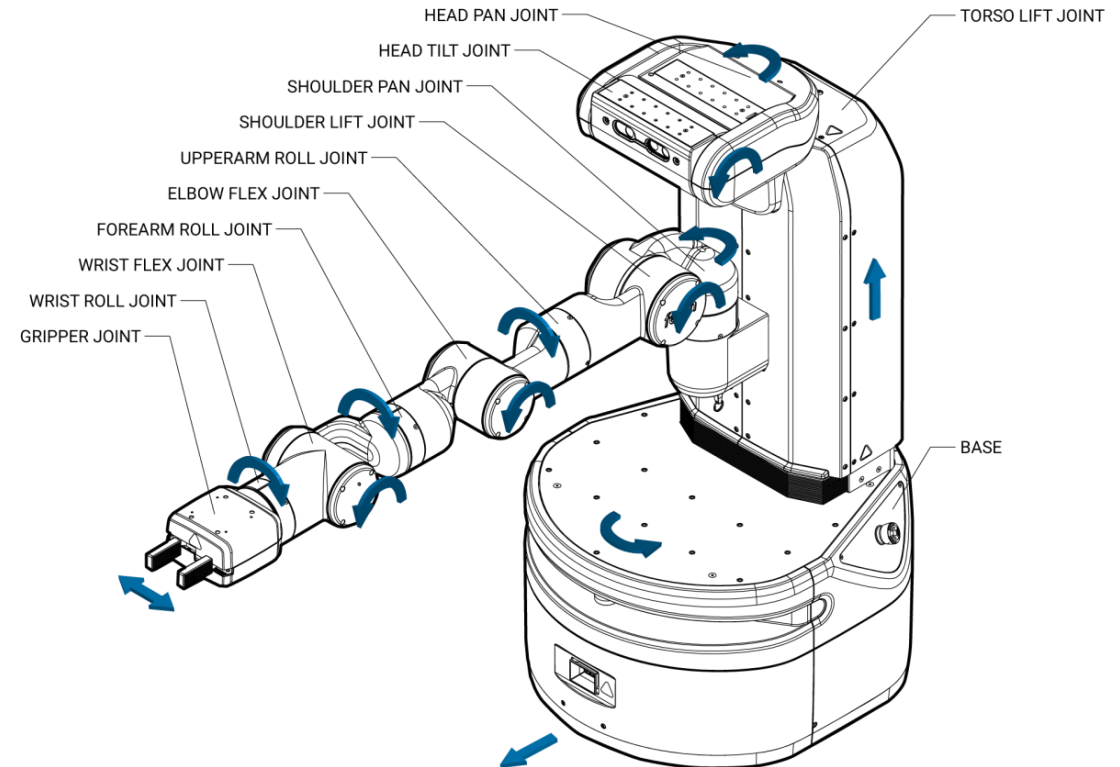
Planar rigid body: 3 DOF

Spatial rigid body: 6 DOF

Degrees of Freedom of a Robot



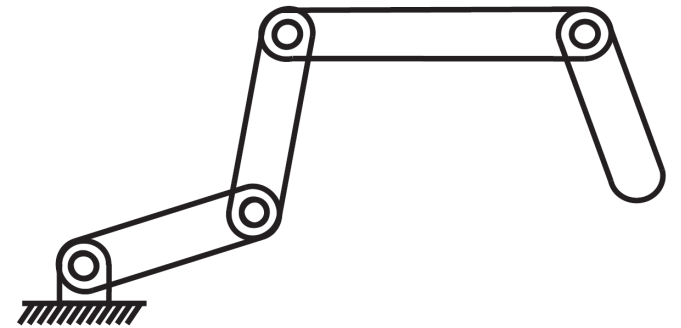
- 4 revolute joints
- 4 DOFs



- 7 revolute joints for the arm
- 7 DOFs

Configuration Space of a Robot

- The configuration of a robot is a complete specification of the position of every point of the robot.
- The minimum number n of real-valued coordinates needed to represent the configuration is the number of degrees of freedom (DOF) of the robot.
- The n -dimensional space containing all possible configurations of the robot is called the configuration space (C-space).
- The configuration of a robot is represented by a point in its C-space.

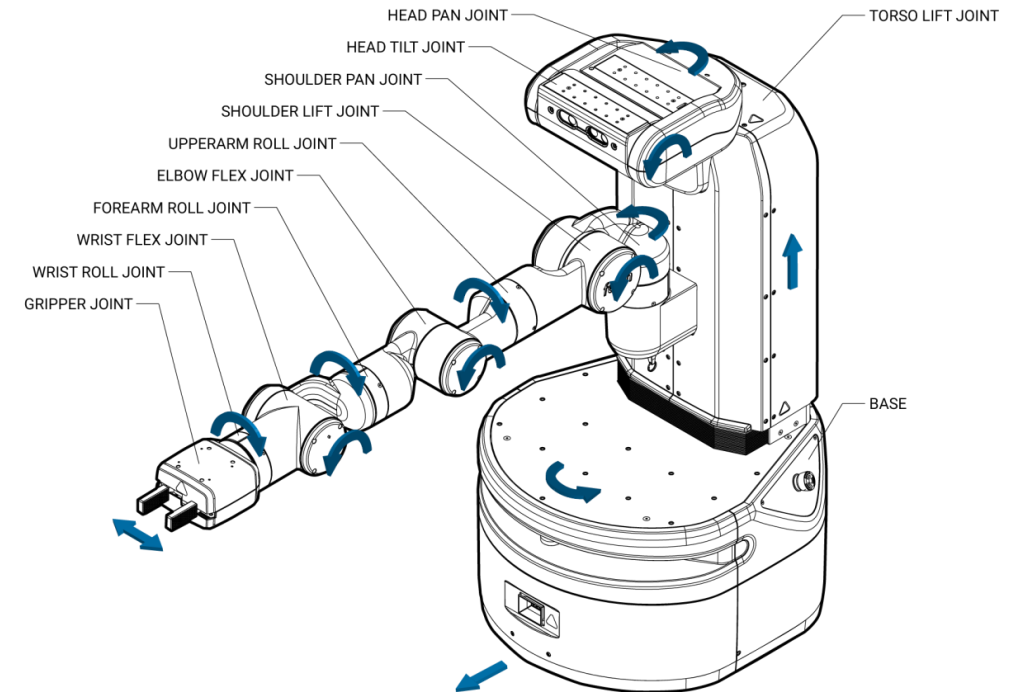


- 4 revolute joints
- 4 DOFs

$$\mathbf{q} \in \mathbb{R}^4$$

Configuration Space of a Robot

- The configuration space of the Fetch arm is a 7D space
- Each value in the 7D vector indicates the value of the revolute joint



Summary

- Robot links and joints
- Degrees of freedom of joints and robots
- Configuration space

Further Reading

- Chapter 2 in Kevin M. Lynch and Frank C. Park. Modern Robotics: Mechanics, Planning, and Control. 1st Edition, 2017
<http://hades.mech.northwestern.edu/images/7/7f/MR.pdf>
- T. Lozano-Perez. Spatial planning: a configuration space approach. A.I. Memo 605, MIT Artificial Intelligence Laboratory, 1980.
<http://people.csail.mit.edu/tlp/>
- W. M. Boothby. An Introduction to Differentiable Manifolds and Riemannian Geometry. Academic Press, 2002.